

	Work Instruction	Doc. Revision	01
	AXI V810 XXL VM (19um and 13um) Tube Height Adjustment Procedure	Effective Date	2 Oct 2016

This procedure is applicable to below models:

V810 STD	☐	S1	☐	S2
V810 XXL	☒	S1	☒	S2
V810 S2EX	☐			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	09 Apr 2015	First Release	CHIN CHEE YAN
01	2 Oct 2016	Template Changes & Content Update	ONG KEAN SHENG

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Low Magnification (19um) Tube Height Adjustment Procedure

Steps:

1. Go to path C:\Program Files\AXI System\v810\<REV>\xxlconfig and open software.config with notepad format.
2. Search for “saveAllConfirmationImages”, set indicator value to “true” and save config. May refer to Figure 1

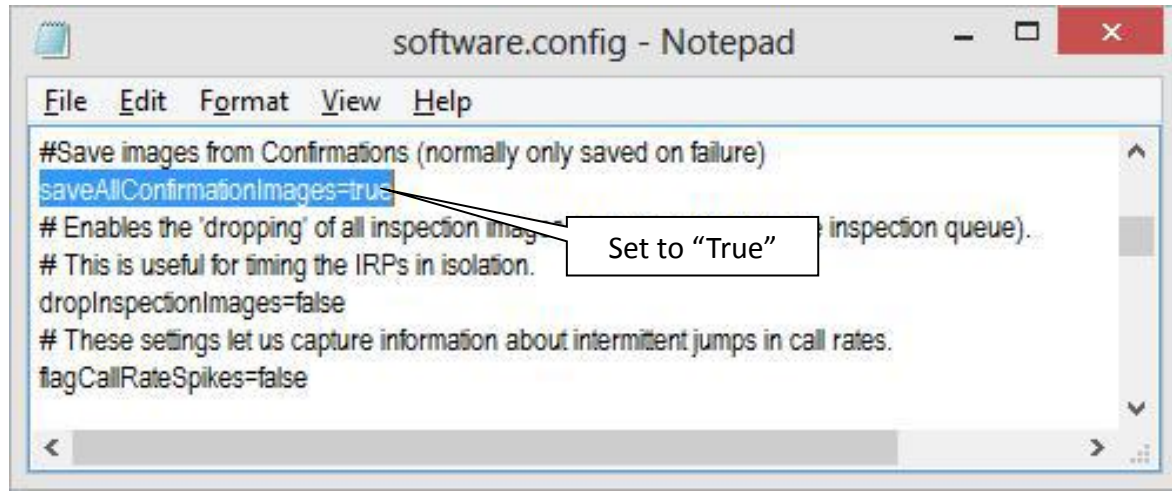


Figure 1

3. Quit V810 GUI and restart V810 GUI as usual.
4. Perform 1 loop full CDnA task to obtain latest log file. Go to Confirmation, Diagnostic and Adjustment tab as Figure 2.

Services → Confirmation, Diagnostic and Adjustment → Scheduled Tasks → Run 1 loop

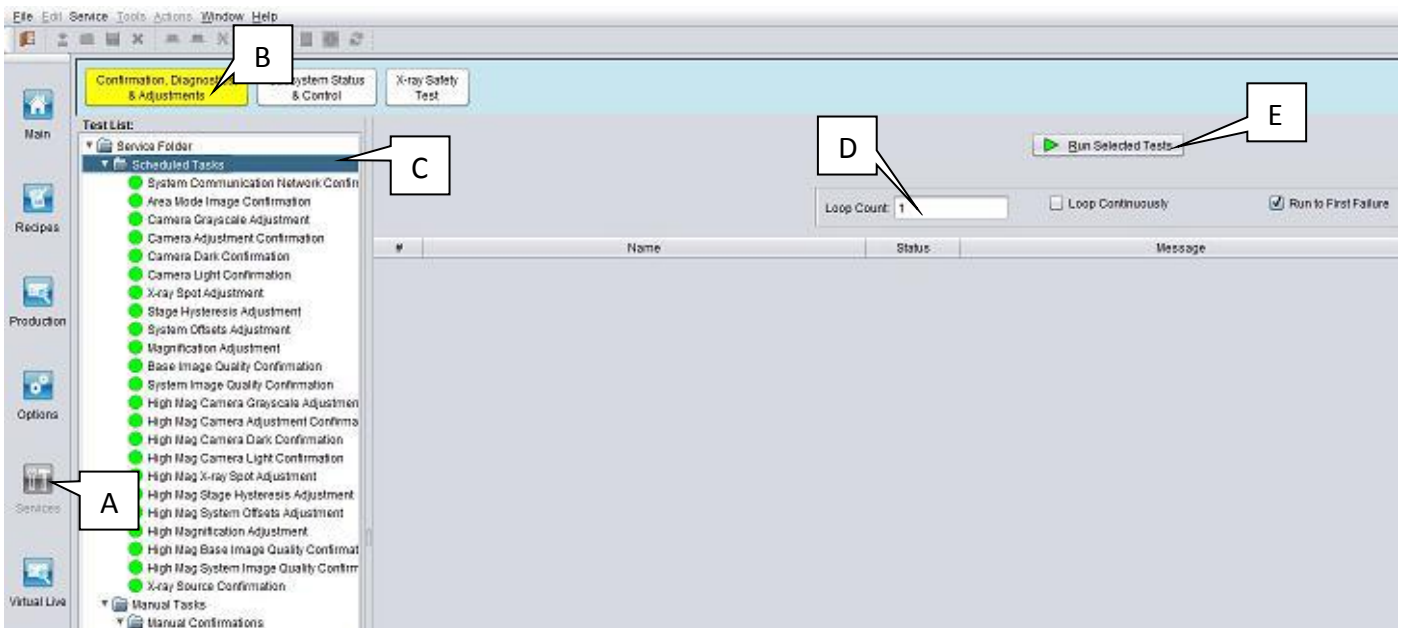


Figure 2

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5. Once CDnA complete, go to path C:\Program Files\AXI System\v810\<REV>\log\calib\Magnification, check for latest low magnification log file. Sample as Figure 3. (# File name indicate as *magnificationYYYY.MM.DD.HH.MM.SS.XXX.log*)

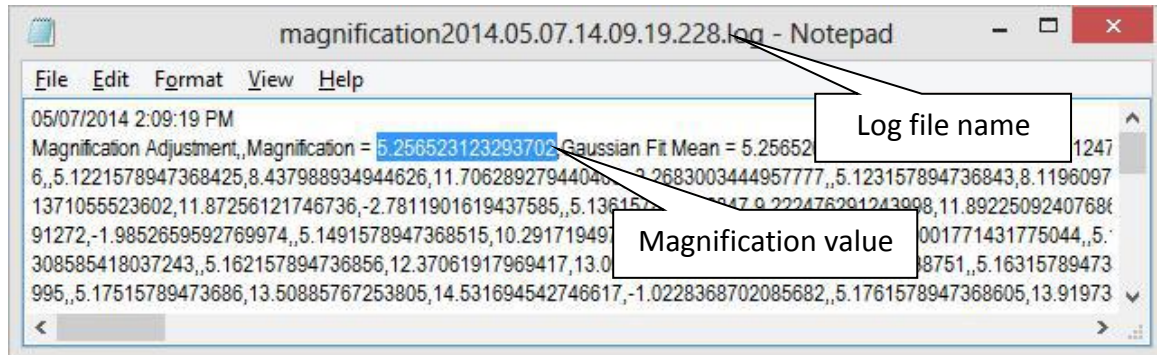


Figure 3

6. From the latest log file, calculate the required adjustment mils need to perform at micrometer with formula below.

$$[A = (5.25 - B) / 0.004]$$

A = mils required

B = latest log files

1 turn of Micrometer = 40 mils

If calculation result is + value, mean you need to lower tube Z position

If calculation result is - value, mean you need to higher tube Z position

Sample

If latest log file is 5.15

$$A = (5.25 - 5.15) / 0.004$$

$$A = 25 \text{ mil}$$

Lower down the tube by turn 25mil to achieve 5.25 tube heights.

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7. Go to Digital I/O tab and initialize as Figure 4.

Services → Subsystem Status & Control → Digital I/O → Initialize

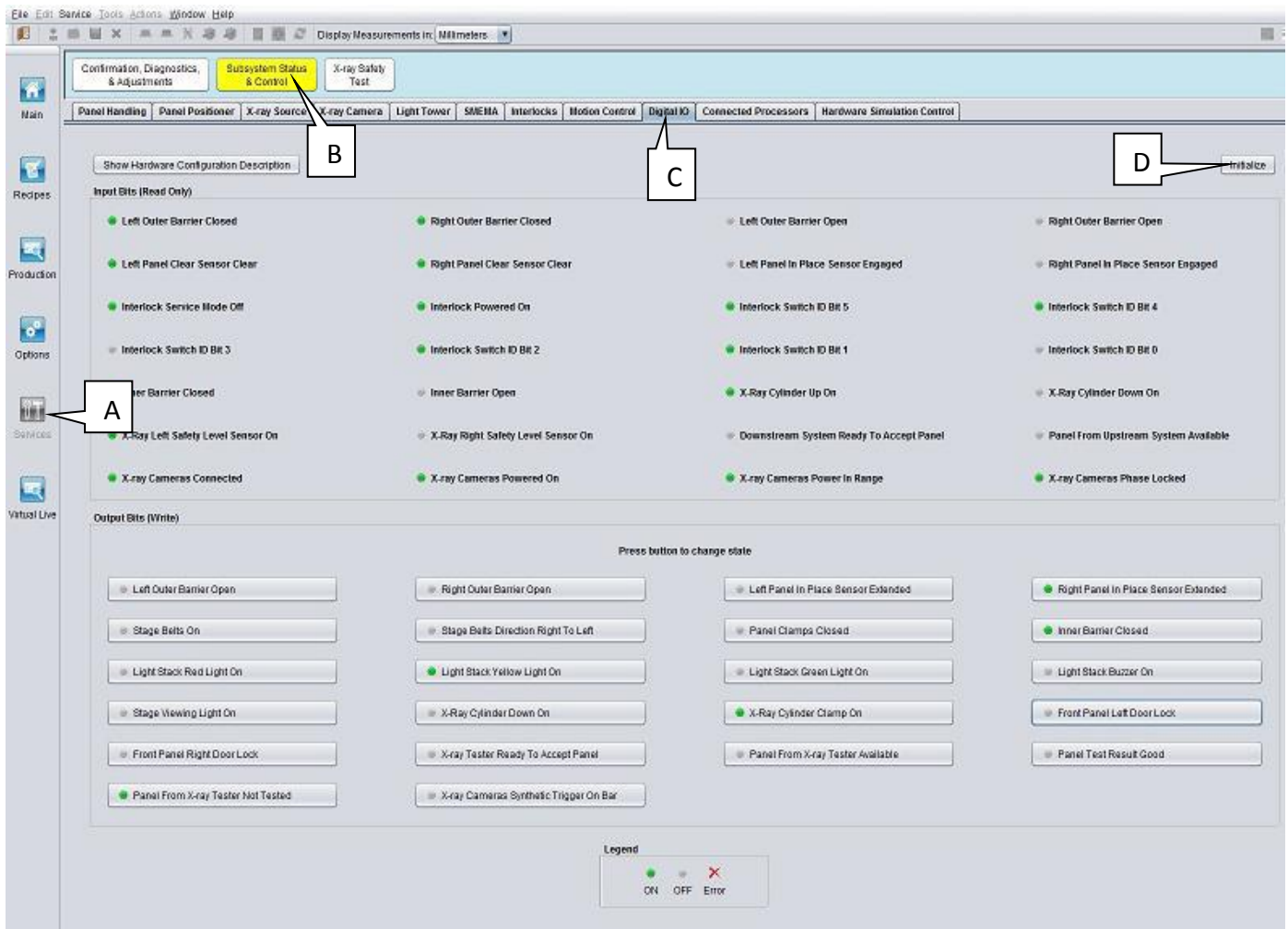


Figure 4

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8. Go to Panel Handling tab and initialize as show in Figure 5. Change display measurement unit to inch and set AWA rail width to 26 inch. Trigger Inner Barrier in Open position, tube cylinder to “Up” position and Clamp in Open position.

Services → Subsystem Status & Control → Panel Handling → display unit to Inch → Initialize → 18 inch width → Inner Barrier Open Position → X-Ray Cylinder Up Position

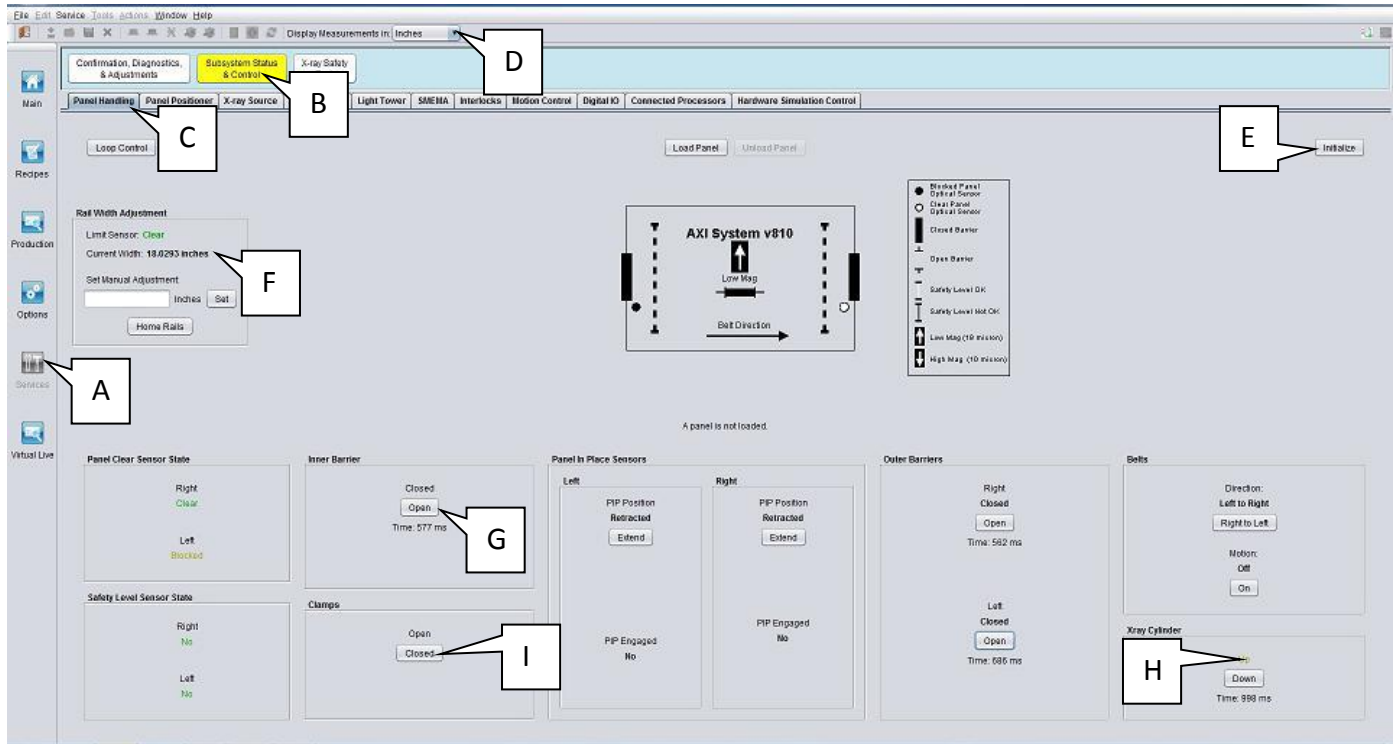


Figure 5

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9. Go to X-Ray Source tab turn off the X-Ray voltage as Figure 6. Then turn off X-Ray Key at Operator Console and open the machine Front Access Door.

Services → Subsystem Status & Control → X-Ray Source → Turn Off Current & Voltage

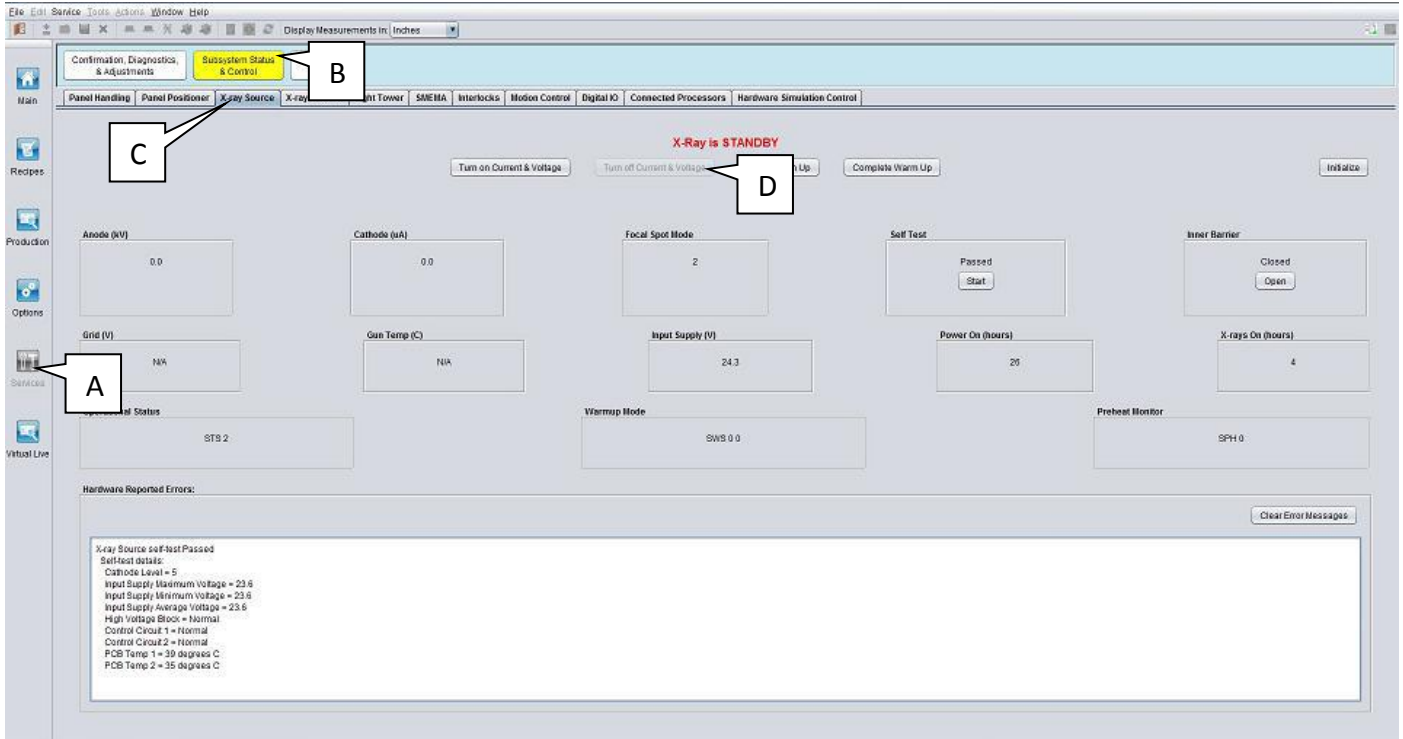


Figure 6

10. Remove the U-Shield Front Cover with loosen 4 x M5 nuts as Figure 7. Left is High Mag hard stop and Right is Low Mag hard stop.

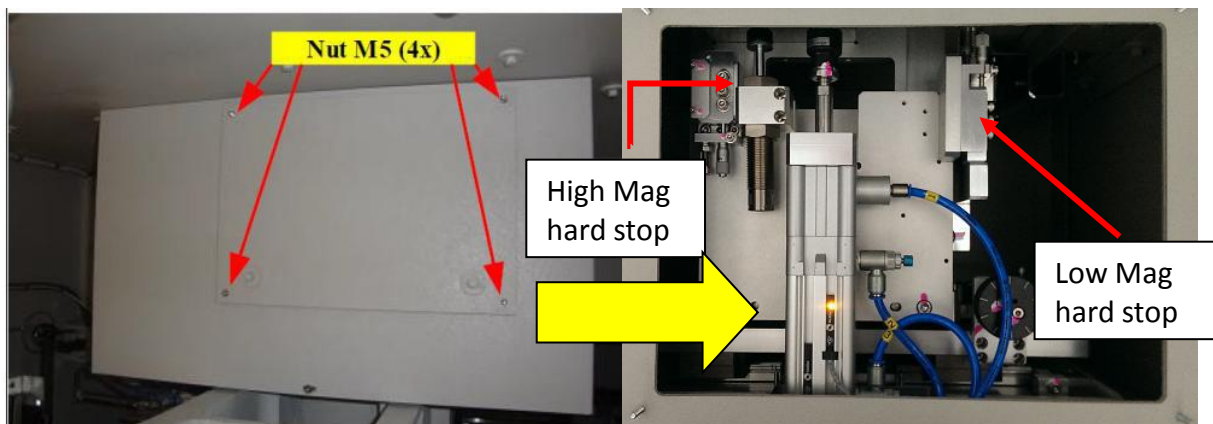


Figure 7

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11. Push the XY stage to middle and ensure cylinder down position will not have any obstacle especially with the Clamper as Figure 8.

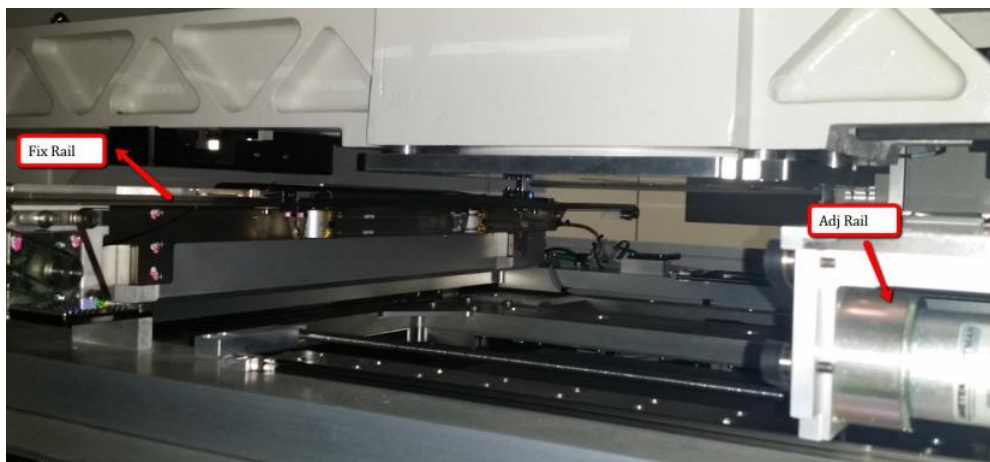


Figure 8

12. Log Mag Hard Stop and screws as Figure 9.

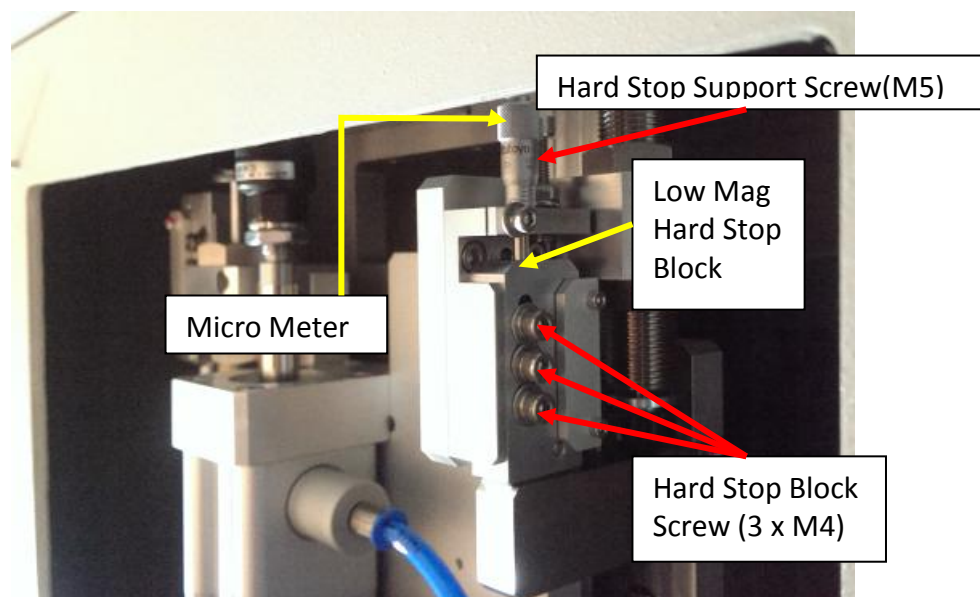


Figure 9 (New Rev)

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Without
Hard Stop
Support
Screw

Figure 10 (Old Rev)

Remark,

If machine assemble with VM Old Rev as Figure 10 will not assemble with Hard Stop Support Screw (M5) ; Tube height adjustment will remain as usual

13. **Remark, If wish to lower tube Z for Low Mag adjustment; require to trigger tube into Down position 1st.**

Go to Panel Handling tab, trigger tube cylinder to “Down” position 1st as Figure 11.

Services → Subsystem Status & Control → Panel Handling → Inner Barrier Open Position → X-Ray Cylinder Down Position

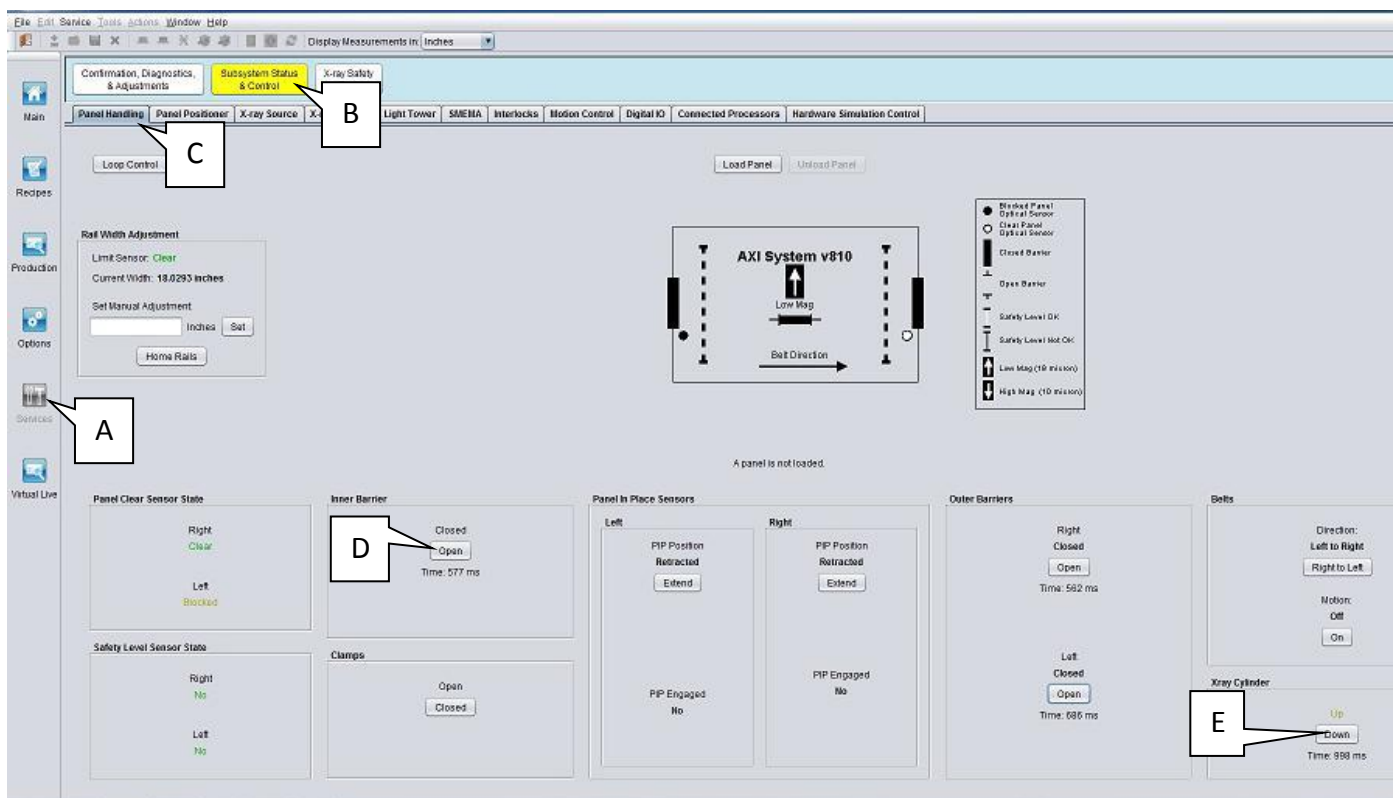


Figure 11

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14. Loosen Hard stop Block Screw (3 x M4) and the Hard Stop Support Screw & Nut (M5) as Figure 12.

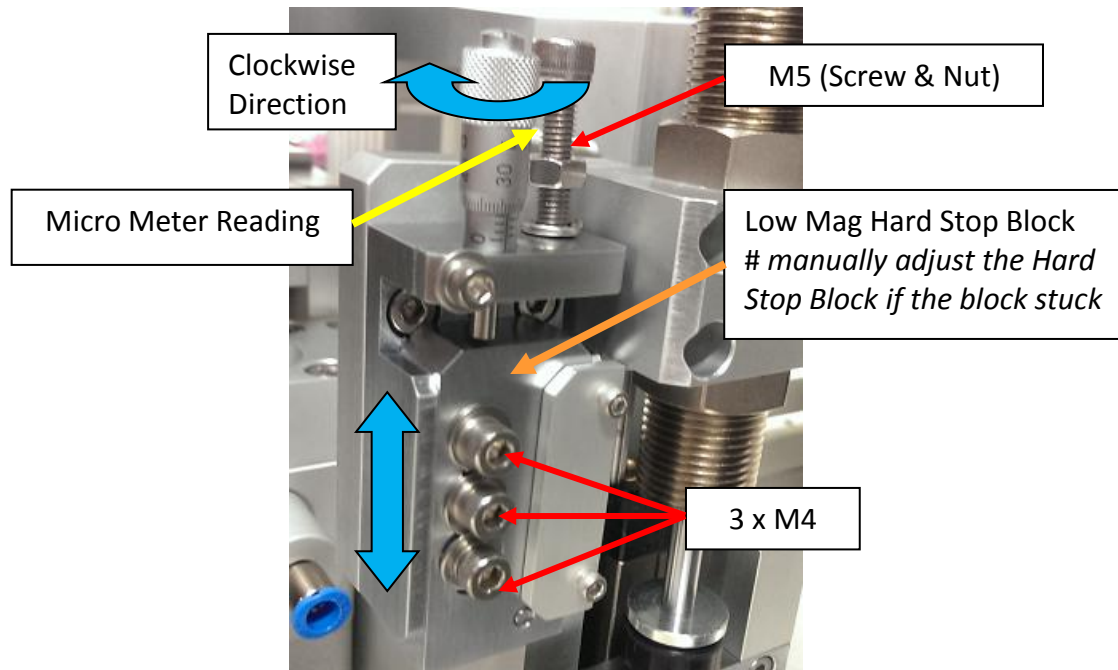


Figure 12

15. Twist Micrometer with **Clockwise (CW)** will lower down the tube and **Counter Clockwise (CCW)** will move the tube to upper position.

Depend on calculation step 6, 1 turn will be 40mil.

May require to manually adjust the Hard Stop Block

Hard Stop block require to contain firm contact with micrometer

*# Spec tolerance **5.21 ~5.25***

16. Tighten 3 x Hard stop Block Screws (M4) once adjustment done.

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17. Go to Panel Handling tab, trigger tube cylinder to “Up” and “Down” position few times (manual exercise) & maintain at Up position before proceed next step.

*Services → Subsystem Status & Control → Digital I/O → Inner Barrier Open Position → X-Ray Cylinder Up/Down Position
→ X-Ray Cylinder Up Position*

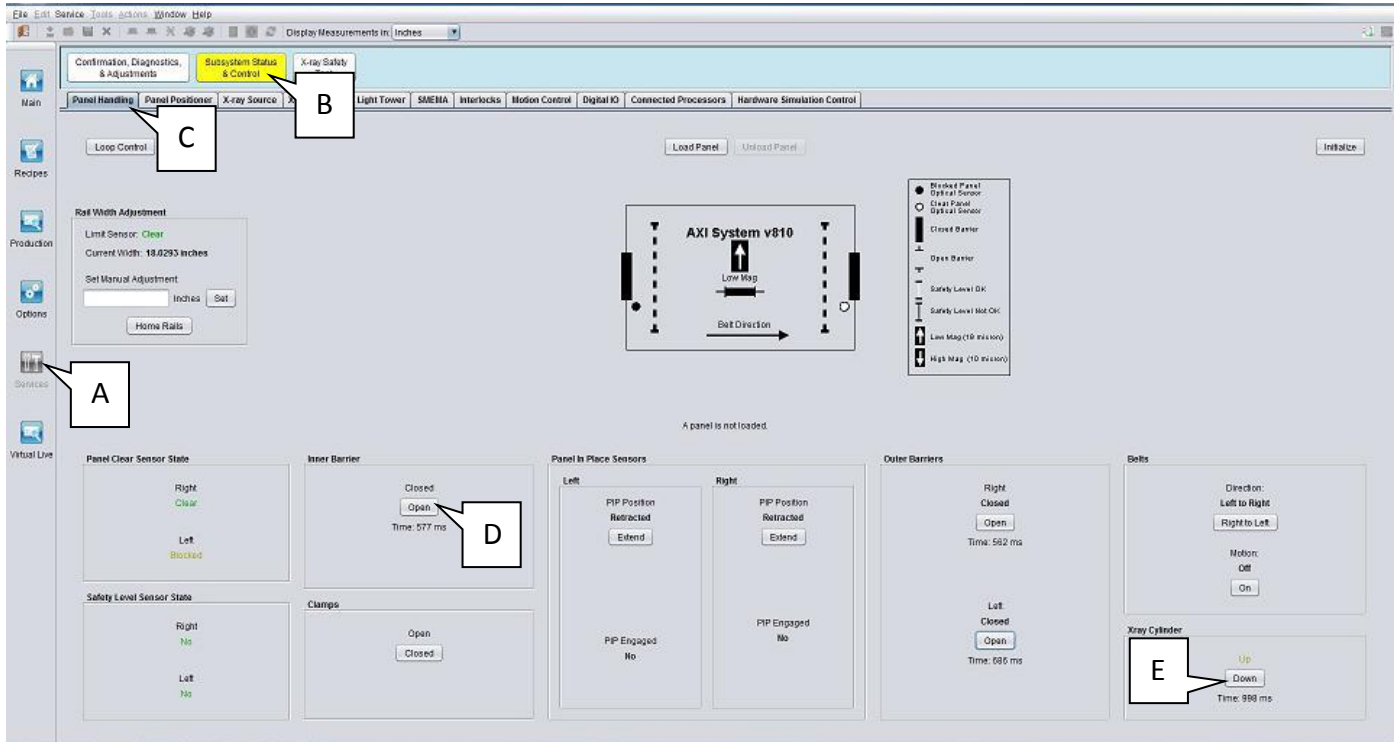


Figure 13

18. Turn off main V810 air pressure regulator.



Figure 14

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19. Manually push the inner barrier shutter plate, ensure X ray tube Beryllium surface is **NOT** Contain friction with Inner Barrier Shutter Plate as Figure 15.

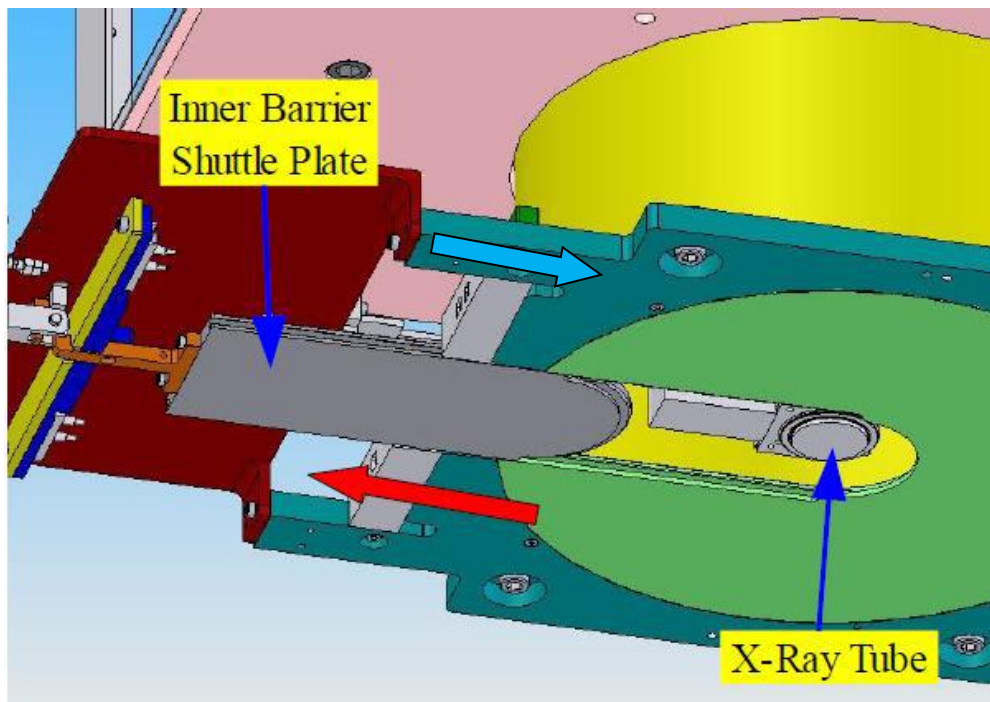


Figure 15

Remark,

Gap between shutter plate and tube Beryllium surface should be slightly as 0.5 mm, as Figure 16. If X-Ray tube target contain friction with Inner barrier shutter plate, trigger cylinder down at Panel Handling and slightly adjust the Low Mag hard stop to higher position.

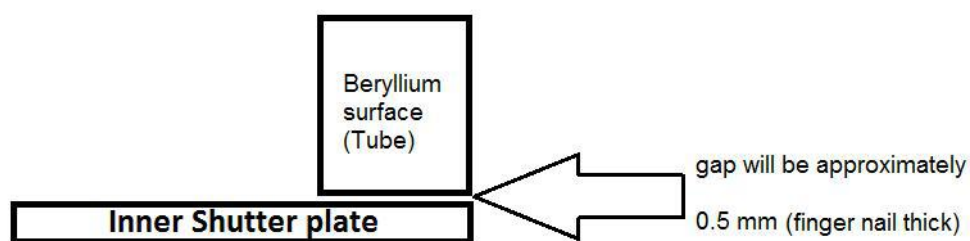


Figure 16

20. Turn On main V810 air pressure regulator and set incoming air source as 80 PSI.

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21. Go to Panel Handling tab, ensure tube cylinder in Up position, manually trigger Inner Barrier Open/Close few times, and check for the shutter plate movement as Figure 17.

Services → Subsystem Status & Control → Panel Handling → X-Ray Cylinder Down Position → Manually trigger Inner Barrier Open/Close

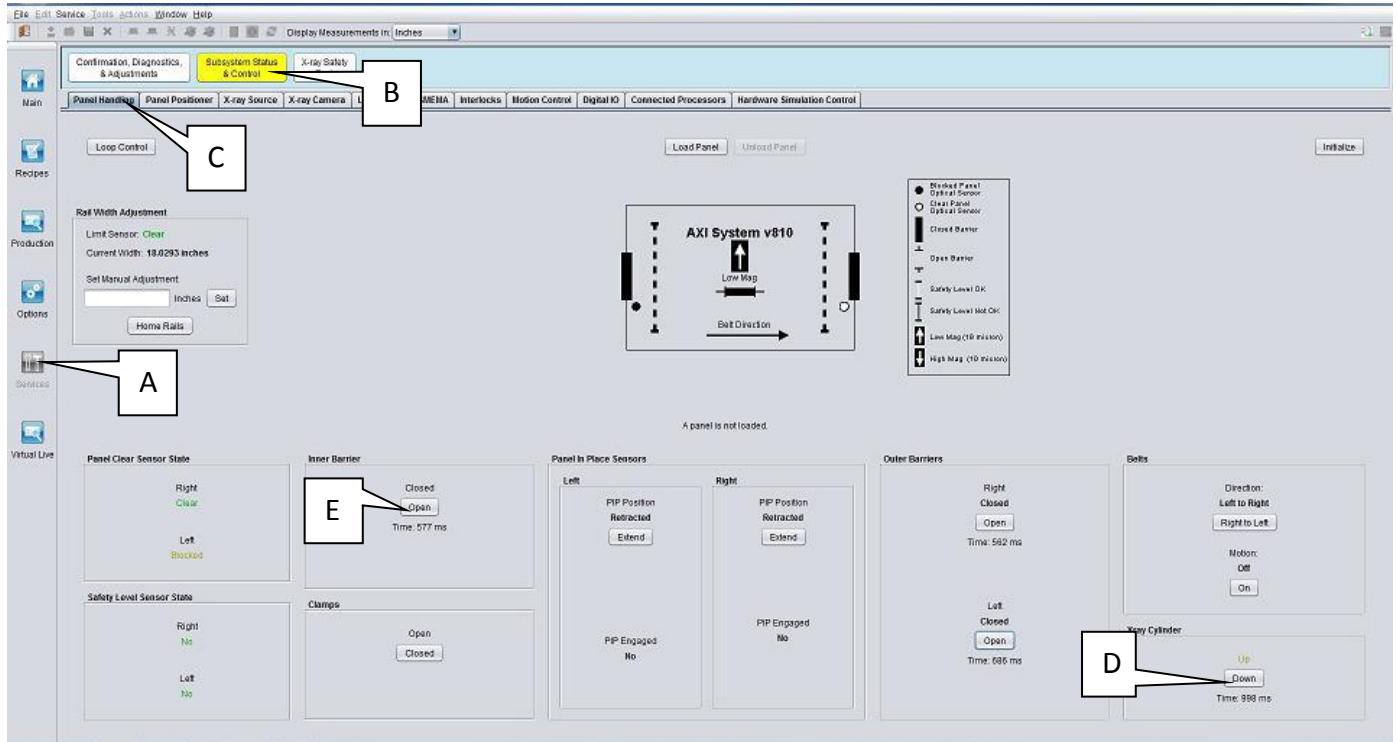


Figure 17

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22. While tube cylinder locate at **Upper** position, hand tight Hard Stop Support Screw & Nut. Ensure the screw is firmly contact with the Hard Stop Block surface as Figure 18.

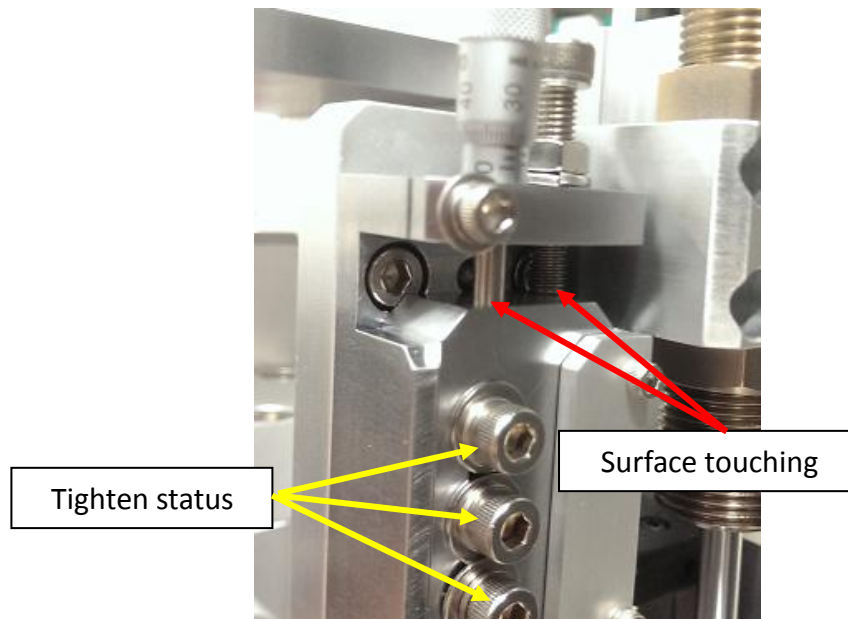


Figure 18

23. While tube cylinder locate at **Upper** position, check proximity sensor signal. If the sensor **NOT** emitted as Figure 19, loosen the set screw and re-position the sensor from top to bottom until LED is turn On (Amber indicate).

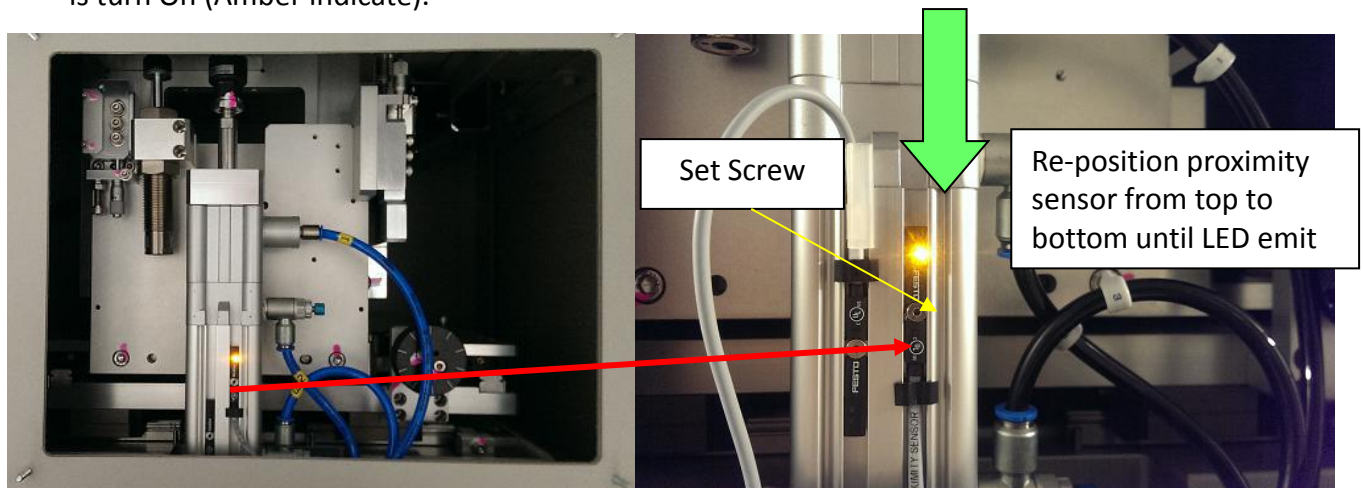


Figure 19

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High Magnification (13um) Tube Height Adjustment Procedure

Steps:

24. Repeat step 1 ~ 11, if directly start with High Mag tube Z adjustment.

Remark on step 5, will be difference log file need to refer to step 25 and calculation for High Mag will refer to step 26.

25. Once CDnA complete, go to path C:\Program Files\AXI System\v810\<REV>\log\calib\Magnification, check for latest high magnification log file. Sample as Figure 20.

(# File name indicate as *highMagnificationYYYY.MM.DD.HH.MM.SS.XXX.log*)

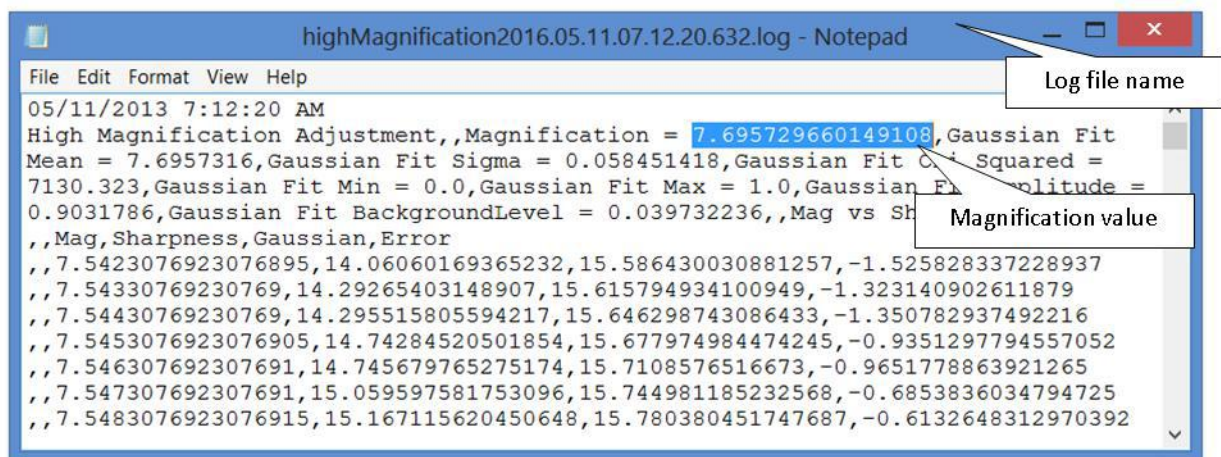


Figure 20

26. From the latest log file, calculate the required adjustment mils need to perform at micrometer with formula below.

$$[A = (7.69 - B) / 0.004]$$

A = mils required

B = latest log files

1 turn of Micrometer = 40 mils

If calculation result is + value, mean you need to lower tube Z position

If calculation result is - value, mean you need to higher tube Z position

Sample

If latest log file is 7.75

$$A = (7.69 - 7.75) / 0.004$$

$$A = -15 \text{ mil}$$

Require to increase tube height by turn 15mil to achieve 7.69 tube heights.

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27. High Mag Hard Stop and screws as Figure 21.

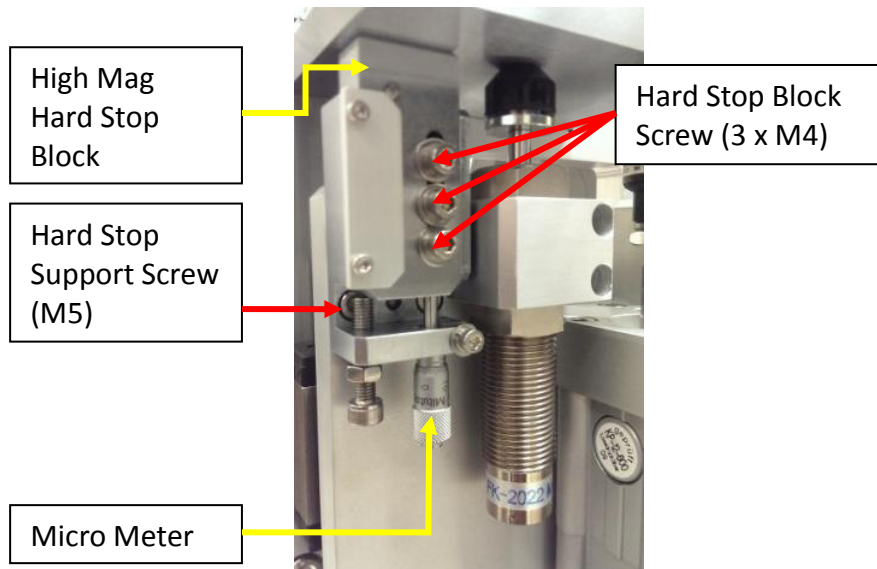


Figure 21 (New Rev)

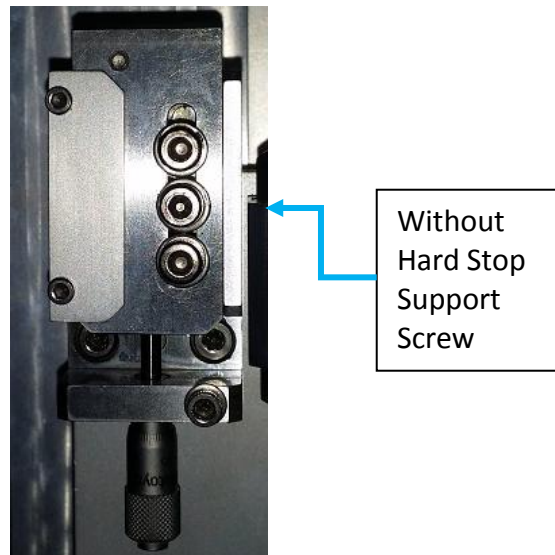


Figure 22 (Old Rev)

Remark,

If machine assemble with VM Old Rev as Figure 22 will not assemble Hard Stop Support Screw (M5), tube adjustment will be the same method.

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28. Remark, If require to move tube to upper position for High Mag adjustment, require to trigger tube in Down position in advance.

Go to Panel Handling tab, trigger tube cylinder to “Down” position 1st as Figure 23.

Services → Subsystem Status & Control → Panel Handling → Inner Barrier Open Position → X-Ray Cylinder Down Position

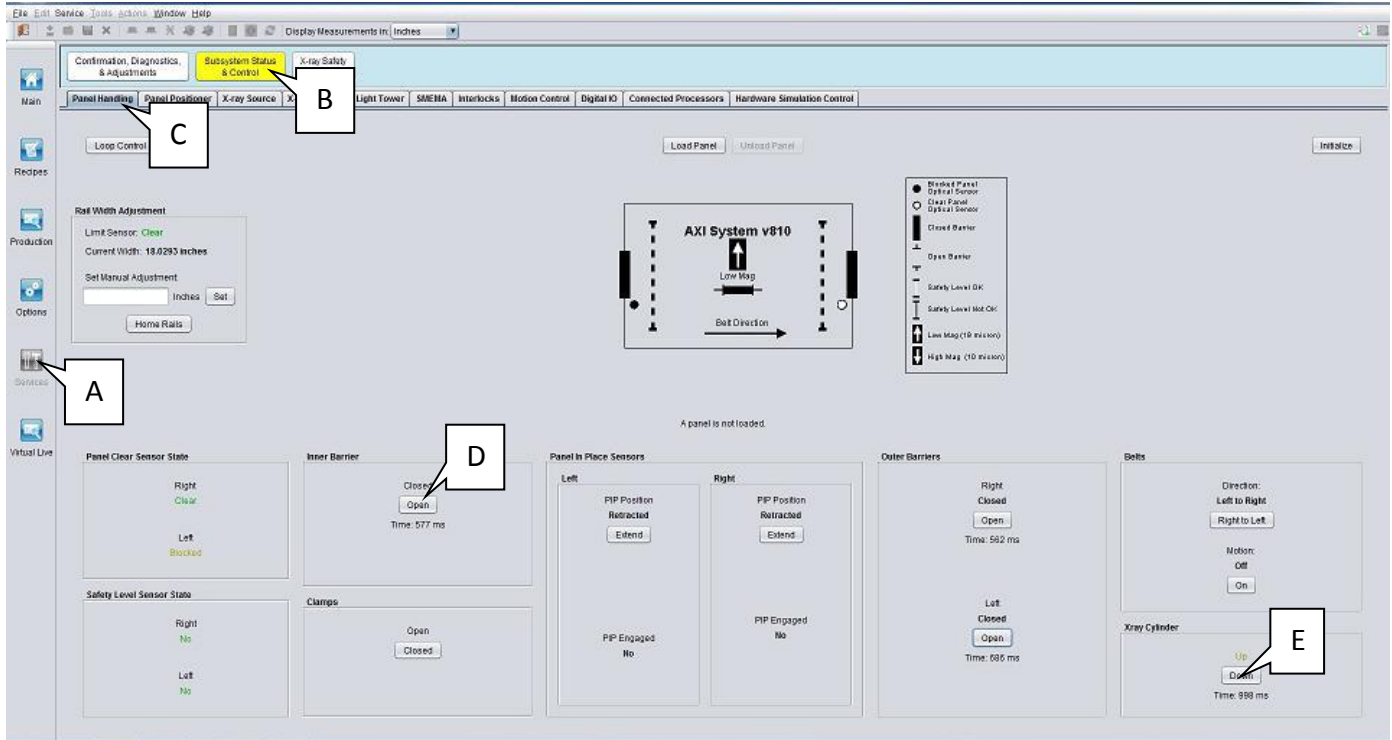


Figure 23

29. Loosen 3 x Hard stop Block Screw (M4) and the Hard Stop Support Screw & Nut (M5) as Figure 24.

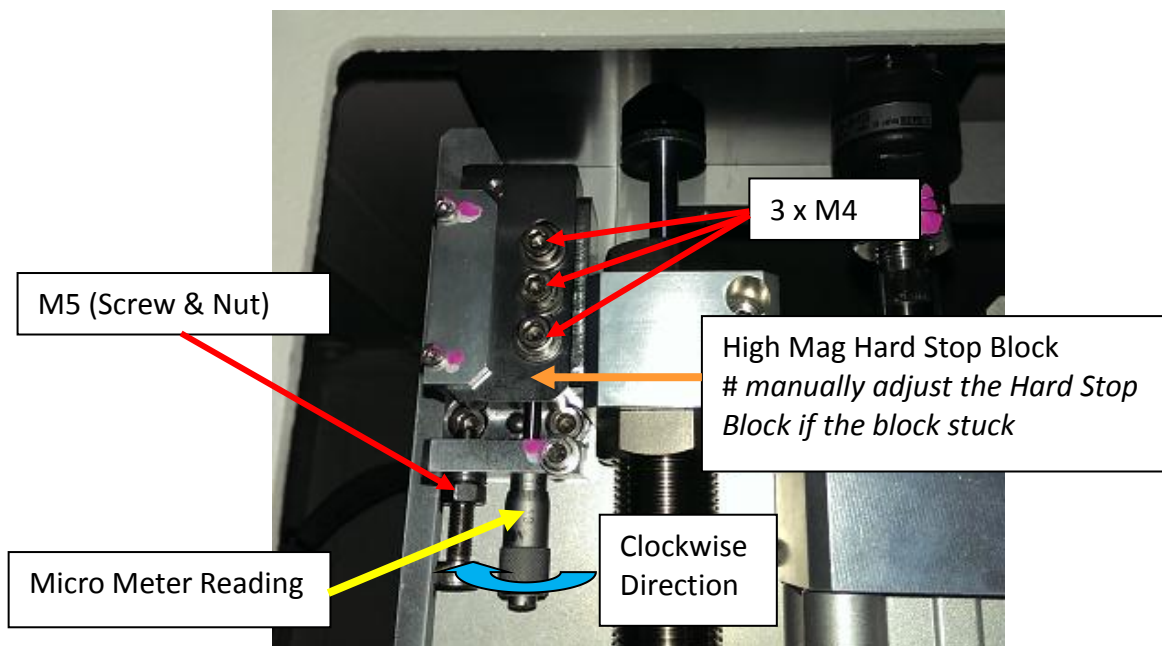


Figure 24

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30. Twist Micrometer with **Clockwise (CW)** will lower the tube and **Counter Clockwise (CCW)** higher the tube.

Depend on calculation step 26, 1 turn will be 40mil.

May require to manually adjust the Hard Stop Block

Hard Stop block require to contain firm contact with micrometer

*# Spec tolerance **7.65 ~ 7.69***

31. Tighten 3 x Hard stop Block Screws (M4) once adjustment done.

32. Go to Panel Handling tab, trigger tube cylinder to “Up” and “Down” position few times (manual exercise) and keep in Down position before proceed next step.

Services → Subsystem Status & Control → Panel Handling → Inner Barrier Open Position → X-Ray Cylinder Up/Down Position → X-Ray Cylinder Down Position

33. Place the Ver Mag Jig (4LASS-A265) on the Rail stage with Top Clearance Block face top as Figure 25.

Trigger “Panel Clamp Closed” at Digital IO to clamp the Ver Mag Jig.



Figure 25

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34. Move the XY stage pass thru the X-Ray tube Beryllium surface; ensure Jig is not crash/friction with tube Figure 26 & stage clumper as Figure 27. As usual the clearance between tube and Jig will be approximately 2~3mm. If X-Ray tube target contain friction with jig, slightly adjust the High Mag hard stop to higher position.

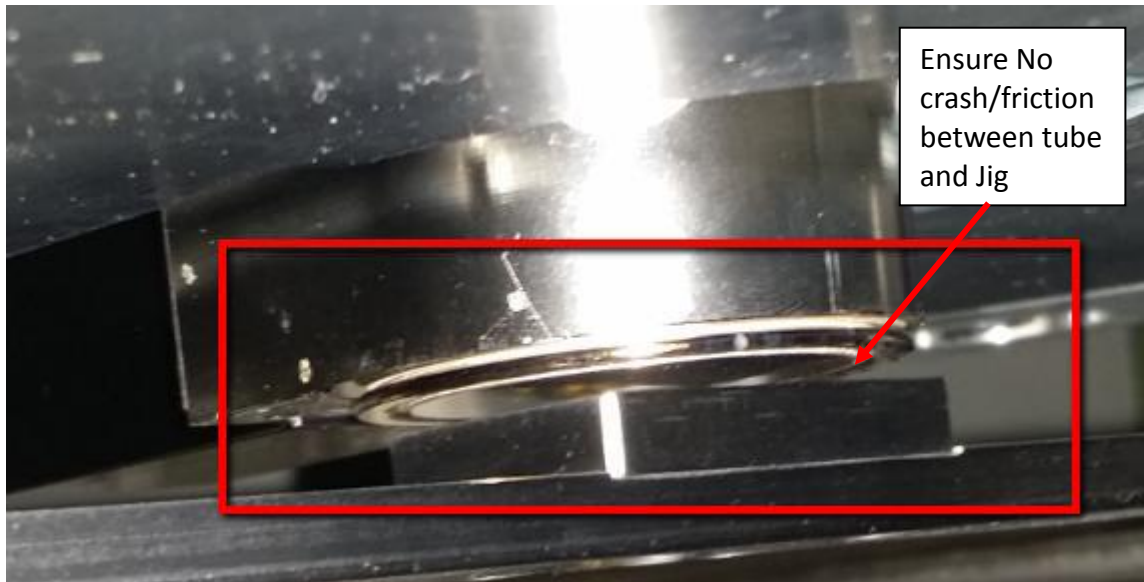


Figure 26

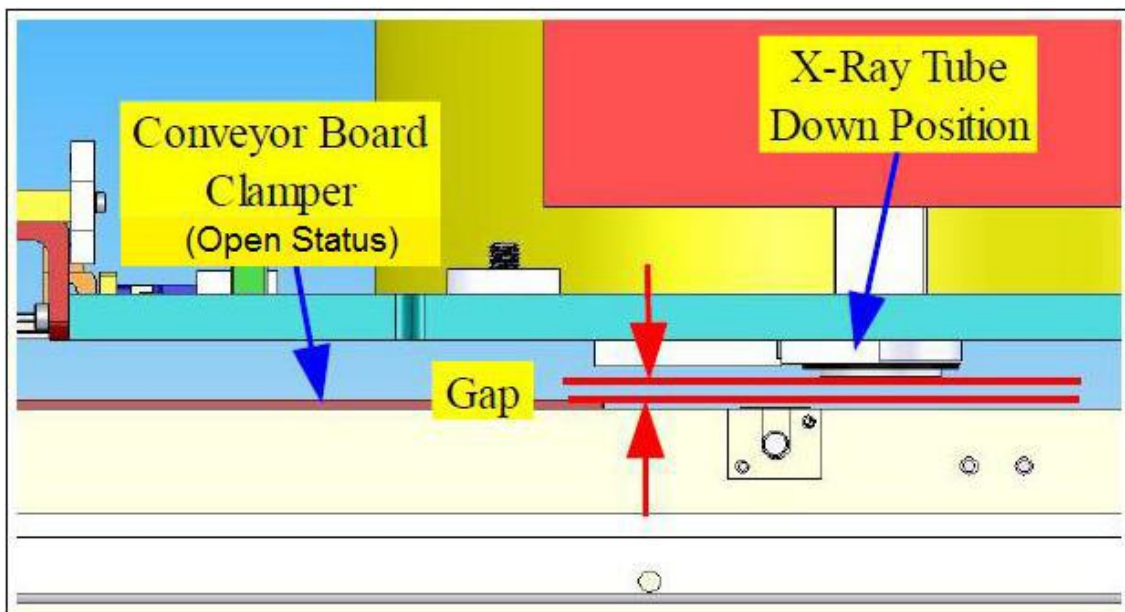


Figure 27

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35. While tube cylinder at **Down** position, hand tight the M5, Hard Stop Support Screw and Nut. Ensure the screw is touching the Hard Stop Block surface as Figure 28.

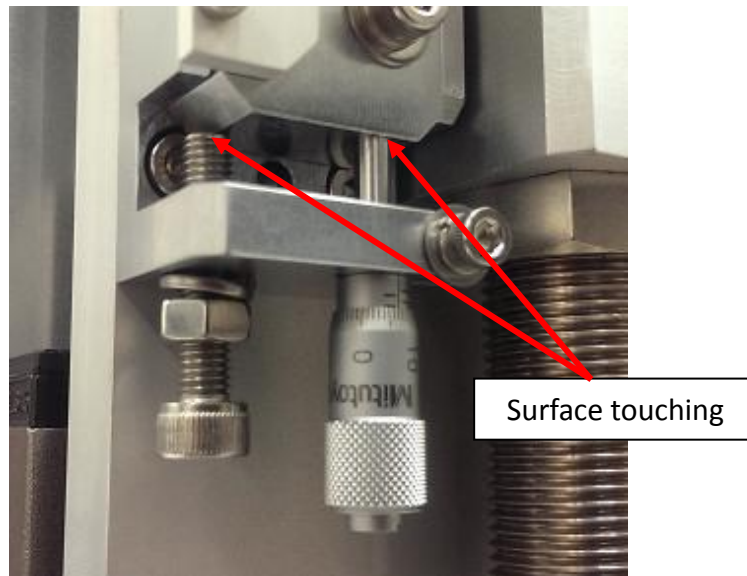


Figure 28

36. While tube cylinder at Down position, check for proximity sensor signal. If the sensor **NOT** emitted, loosen the set screw and reposition the sensor from bottom to top until LED is turn On (Amber indicate).

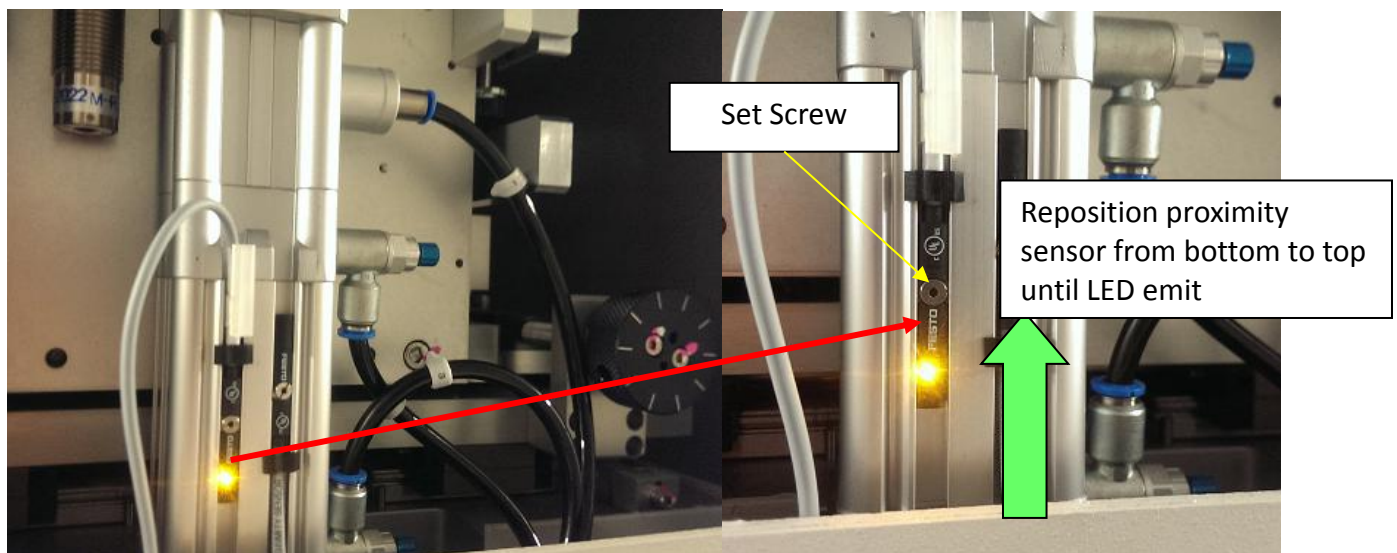


Figure 29

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37. Go to Panel Handling tab, trigger tube cylinder to “Up” position as Figure 30.

Services → Subsystem Status & Control → Panel Handling → Inner Barrier Open Position → X-Ray Cylinder Up Position.

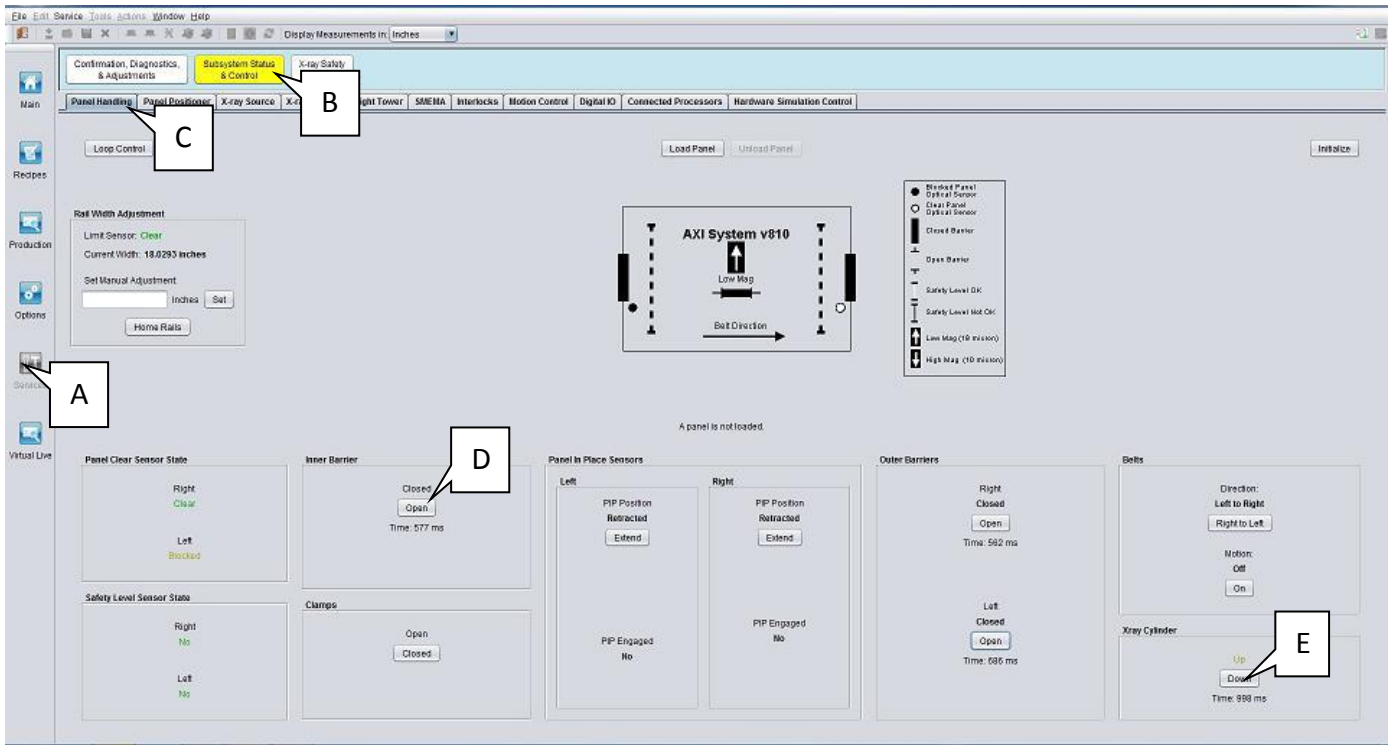


Figure 30

38. Assemble U-Shield Front Cover with fastener 4 x M5 nuts.

39. Remove all unnecessary tools and jig from machine.

Start-up machine as usual.

40. Run full CDnA 5 loop and verify each loop magnification log file (Low and High Mag) is within below spec.

*Low Mag Spec tolerance **5.20~5.25***

*High Mag Spec tolerance **7.65~7.69***